

Get count data from STAR output

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```
# Set CRAN mirror explicitly
options(repos = c(CRAN = "https://cloud.r-project.org"))

knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE)

library(tidyverse)
library(dplyr)
library(purrr)
library(DESeq2)
library(readr)
library(stringr)
library(purrr)
library(tibble)
library(rtracklayer)
library(tibble)
library(readr)

# Path with STAR outputs
in_dir <- "/home/chu25/data/star/count_data"
out_dir <- "/home/chu25/data/dsmz"
if (!dir.exists(out_dir)) dir.create(out_dir, recursive = TRUE)

# Find files
count_files <- list.files(in_dir, pattern = "ReadsPerGene\\.out\\.tab$", full.names = TRUE)

# Sample names from file names (remove the suffix, anchored)
sample_names <- basename(count_files) %>%
  str_remove("\\.ReadsPerGene\\.out\\.tab$")

# Read each file and name count column by its sample
# Keep col1 (Ensembl with version) and col2 (unstranded counts)
count_list <- purrr::map2(count_files, sample_names, ~{
  df <- readr::read_tsv(.x, col_names = FALSE, show_col_types = FALSE)[, c(1, 2)]
  df <- dplyr::filter(df, !grepl("^N_", .data$X1)) # drop STAR summary rows
  colnames(df) <- c("gene_id_ver", .y) # name columns: gene_id + sample_name
  df
})

# Merge by Ensembl_id
counts <- purrr::reduce(count_list, dplyr::full_join, by = "gene_id_ver")
```

```
# Replace any NAs (if gene missing in some file) with 0
counts[is.na(counts)] <- 0

# Save raw merged counts (gene_id with version as first column)
saveRDS(counts, file = file.path(out_dir, "DSMZ_star2count.rds"))
```

Extract gene_id → gene_name from GENCODE GTF

Load GTF and extract mapping

```
# Import GTF and extract (gene_id, gene_name)
gtf_path <- "/home/chu25/data/GTF/Homo_sapiens.GRCh38.107.gtf"
gtf <- rtracklayer::import(gtf_path)
genes_gtf <- gtf[gtf$type == "gene"]

gene_map <- as.data.frame(mcols(genes_gtf)) %>%
  dplyr::select(gene_id, gene_name) %>%
  dplyr::distinct(gene_id, .keep_all = TRUE) # ensure unique mapping per gene_id

# Add version-less Ensembl ID
counts_ensembl <- counts %>%
  dplyr::mutate(Ensembl_ID = sub("\\\\.\\d+$", "", .data$gene_id_ver))

# Join gene symbols
counts_gene <- counts_ensembl %>%
  dplyr::left_join(gene_map, by = c("Ensembl_ID" = "gene_id"))

# Reorder columns for clarity: Ensembl_ID_with_version, Ensembl_ID, gene_name, <samples...>
counts_final <- counts_gene %>%
  dplyr::relocate(gene_id_ver, Ensembl_ID, gene_name) %>%
  dplyr::rename(Ensembl_ID_with_version = gene_id_ver)

# (Optional) set rownames to Ensembl_ID_with_version and keep a pure numeric matrix after metadata
counts_final_rownames <- counts_final %>%
  tibble::column_to_rownames("Ensembl_ID_with_version")

# Save
saveRDS(counts_final, file = file.path(out_dir, "DSMZ_count_gene.rds"))

# Quick sanity prints
cat("Rows (genes):", nrow(counts_final), "\n")
```

```
## Rows (genes): 60660
```

```
cat("Columns (including 2 Ensembl cols + gene_name):", ncol(counts_final), "\n")
```

```
## Columns (including 2 Ensembl cols + gene_name): 192
```

```
cat("Example sample columns:\n")
```

```
## Example sample columns:
```

```
print(head(colnames(counts_final)[4:min(10, ncol(counts_final))]))
```

```
## [1] "NALM_6" "NG-13640_697_lib227469_5697_7"  
## [3] "NG-13640_BC3_lib236396_5774_5" "NG-13640_BCBL_1_lib241208_5808_6"  
## [5] "NG-13640_BJAB_lib227474_5697_7" "NG-13640_BONNA12_lib236392_5774_5"
```